K-Homology Notes

1 Chapter 1

Some Definitions: We say that a representation of a C^* algebra

$$\pi : A \to B(H)$$

is:

1. Faithful: π is injective
2. Nondegenerate: $\pi(1) = 1$
3. Ample: No operator of A acts as a compact operator.

Setting: Let H be a complex Hilbert space equipped with a hermitian inner product. We denote by $B(H)$ the subalgebra of bounded operators. The *adjoint* T^* of an operator T is defined by $\langle Tv, w\rangle = \langle v, T^*w\rangle$ for all v, w. The adjoint is an involution. Here we also have the C^* identity

$$\|T^*T\| = \|T\|^2.$$

Example. Any $*$ subalgebra of $B(H)$ is a C^* algebra. Conversely, a Banach-$*$ algebra with the C^* identity is called a C^* algebra.

Theorem (Gelfand)**.** *If T is normal, then*

$$\varphi : C^*(T) \to C(\mathit{Spec}(T))$$

is an isomorphism.

Definition. The *dual* of a Banach algebra A is defined to be

$$\hat{A} = \{f : A \to \mathbb{C} : f \text{ a nonzero continuous homomorphism}\}.$$

2 Meeting 2

Some Notes by Chenxi:

1. Functional calculus can be viewed as $C(X)$ acting on $L^2(X, \mu)$.

   Fact: We can write
   $$H = \sum_v \operatorname{span}(v, Tv, \dots)$$
   where the v are obtained using Zorn's lemma.

   Fact: We have $\langle P(T)v, v\rangle$ for any polynomial P, and we use Riesz to extend to arbitrary continuous f.

   Then we have a functional on H, so there is a Borel measure μ such that
   $$\langle f(T)v, v\rangle = \int_X f(x)d\mu$$

   Unbounded things: Extend using mobius transform.

Notes by Lily:

Definition. We define the **Caulkin algebra** to be the quotient $B(H)/K(H)$, where B and K denote the spaces of bounded or compact operators.

The **Fredholm Index**, $\operatorname{Ind}(T) = \dim\ker(T) - \dim\operatorname{coker}(T)$.

Theorem (Atkinson)**.** *$T \in B(H)$ is fredholm iff its image in the Calkin algebra is invertible.*

We also have $\operatorname{Ind}(ST) = \operatorname{Ind}(S) + \operatorname{Ind}(T)$.

Definition. Let $T \in B(H)$. The **Essential Spectrum** of T is defined as the ordinary spectrum of the image of T in the Calkin algebra.